Whole Wheat Rolls, Bread Sticks, Buns, 51%

Breads **HACCP: Non-Hazardous/Other** **Healthier Kansas Recipe 877**

USDA B-16, modified

Ingredients	100 Servings		____Servings		Directions
	Weight	Measure	Weight	Measure	
Flour, White Whole Wheat Flour, All-purpose, enriched Dry Milk, instant, nonfat* Sugar, granulated Yeast, *instant*, dry Salt	3 lbs + 12 oz 3 lbs + 8 oz 7 oz 11.5 oz 3 oz 1.75 oz	 1¾ cup 1¼ cup + 3 Tbsp ½ cup 2 Tbsp + 2 tsp			1. Place flour, dry milk, sugar, yeast and salt in mixer bowl. 2. Blend with dough hook for approximately 2 minutes on low speed.
Vegetable Oil	13 oz	$1^{2}/_{3}$ cup			3. Add oil and blend for approximately 2 minutes on low speed.
Water* (Calculate water temperature by subtracting the temperature of the dry ingredients from 145°)		2 qt + ¼ cup (+ up to 1 cup if dough is stiff)			4. Add first amount of water to the dry ingredients. If dough appears too stiff, add up to 1 cup extra water per 100 servings. More or less water can be added as needed. Mix for 1 minute on low speed or until all water is mixed with the dry ingredients. 5. Knead dough on medium speed for 8-10 minutes until dough is properly developed. Use the gluten stretch test: At 8 minutes, stretch the dough like a rope. If it breaks, continue to mix 2 additional minutes.
Pan release spray		As needed			6. Form rolls from dough by pinching off 2 oz pieces and shaping. Place rolls in rows of 7 across and 10 down on sheet pans (18" x 26" x 1") which have been lined with pan liners. (See variations below for shaping bread sticks and hamburger buns.) 7. Place in a warm area (about 90° F) until double in size, approximately 45-60 minutes.

Whole Wheat Rolls, Bread Sticks, Buns, 51%, continued

Ingredients	100 Servings		___ Servings		Directions
	Weight	Measure	Weight	Measure	
					8. Bake until lightly browned. * Conventional oven: 400° F for 18-20 minute * Convection oven: 350° F for 12-14 minutes Turn pans half-way through the baking time to promote even baking. Recommended internal temperature for baked rolls is 196-198°F. 9. Rolls will have a better appearance if lightly sprayed with pan release spray when they come out of the oven.

Serving Size	1 Serving Provides	Yield
K-12: 1 each	2.2 units G/B	100 rolls – Approximately 13 lbs 8 oz of dough

Nutrients Per Serving

Calories	160	Vitamin A	9.6 IU	Iron	1.68 mg
Protein	4.02 gm	Vitamin C	2.30 mg	Calcium	17.08 mg
Carbohydrate	28.07 gm	Fiber	2.40 gm	Cholesterol	0 mg
Fat	4.12 gm	% Fat	23.15 %	Sodium	192 mg
Saturated Fat	0.63 gm	% Saturated Fat	3.55 %		

***IMPORTANT!!!**

If applying for the HealthierUS School Challenge Gold Award,
replace instant nonfat dry milk and 2 cups of water with 2 cups of 1% fluid milk.

Bread Stick Variation

1. Follow steps 1 – 5 for Whole Wheat Rolls.

2. At step 6, pinch 2-ounce dough balls stretching to make “sticks”. Place in rows 3 across and 17 down on sheet pans lined with pan liners. OR weigh out 6-pound 6-ounce dough balls. Place one ball per pan. Roll or spread to fill pan. Cut 3 across and 17 down.

3. Melt 10 oz margarine and mix in 1 tablespoon of garlic powder for every 100 breadsticks. Brush top of breadsticks.

4. Continue with remaining steps.

Hamburger Bun Variation

1. Follow steps 1 – 5 above.

2. At step 6, form 1.5 or 2 oz dough balls.

3. Flatten with a rolling pin, or with hands, or place on sheet pans, cover with a tray liner and flatten buns with a second sheet pan.

4. Continue with remaining steps.

Note:
Dough made with whole wheat flour will absorb more water and requires increased rising/proofing time. Mixing time is less because the bran from the whole grain cuts through developing gluten strands with increased mixing resulting in a product with low volume. Even an additional one minute mixing time with whole grains can make a difference in quality. Figuring water temperature so that the final dough temperature is approximately 80°F is essential for quality bread products.